

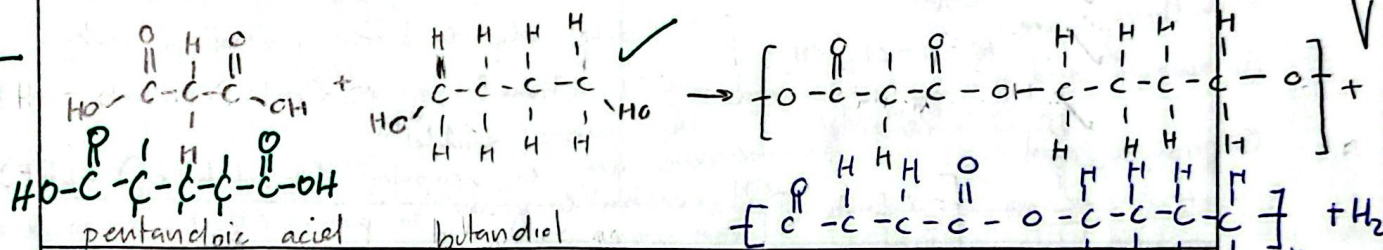
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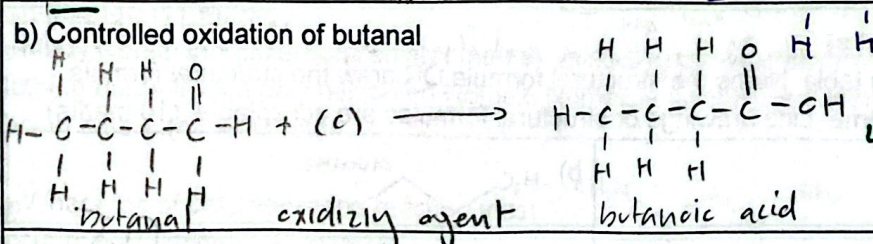
Unit 2 Test: Organic Chemistry**Knowledge (15 marks) – Multiple Choice on D2L****Thinking (14 marks)**

1. Write a chemical equation, using structural formula or carbon skeleton, for each of the following reactions and **name the reactants and products**: (10 Marks total - 2 marks for each reaction)

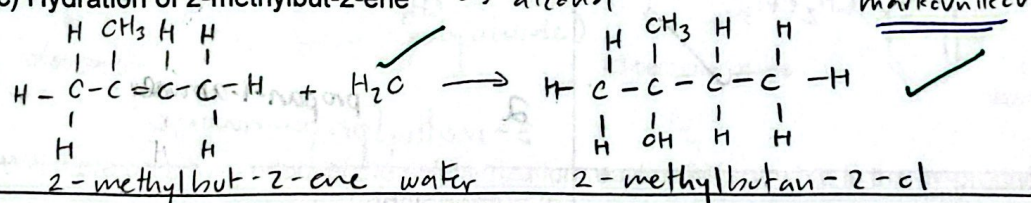
a) Polymerization of pentanedioic acid and butanediol – 2 repeating units (**No naming** – just show structural drawings)



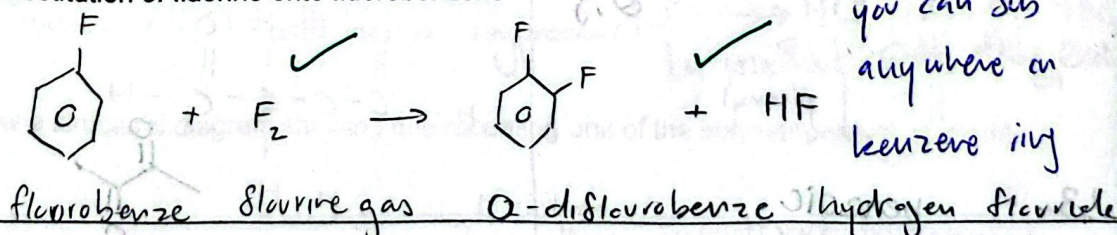
b) Controlled oxidation of butanal



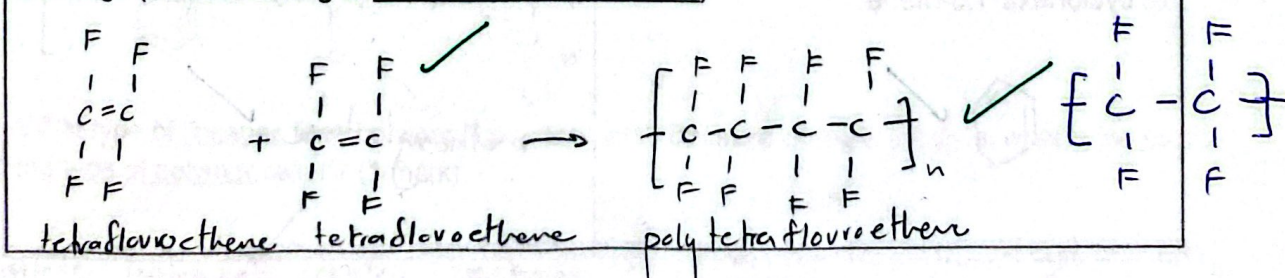
c) Hydration of 2-methylbut-2-ene → alcohol



d) Substitution of fluorine onto fluorobenzene



e) Addition polymerization of tetrafluoroethene (Teflon) – 2 repeating units of structural drawings (include naming of final product and monomer)

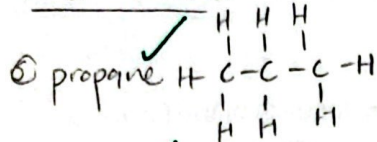


- * the more polar a molecule is, or if it has H-bonds the more soluble it is in water as water is polar (like dissolves like)
2. Rank the following compounds in order of increasing solubility in water. Include a brief explanation. (4 marks)

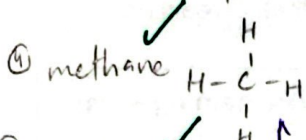
③ ① ④ ② ⑤
Propane, propanoic acid, methane, benzoic acid, methoxyethane

water → polar

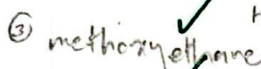
least soluble



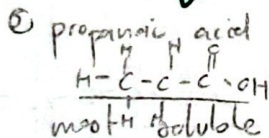
- only LDF forces → non-polar, bad solubility in water
- no functional groups
- low solubility bc larger molecule (larger than methane)



- non-polar, bad solubility in water (polar)
- only LDF, no functional groups



smaller



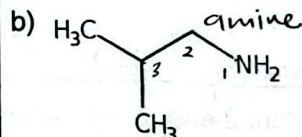
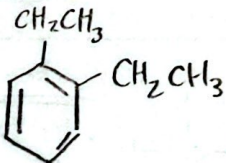
- dipole-dipole forces (mod polarity)
- O functional group (polar) allows it to be more soluble
- benzene ring is non-polar (↓ solubility) (LDF)
- CH functional group is polar (H-bonding = ↑ solubility)
- 2 functional groups = O and OH

Communication (10 marks)

allow for H-bonding and dipole-dipole ↑ polarity = ↑ solubility

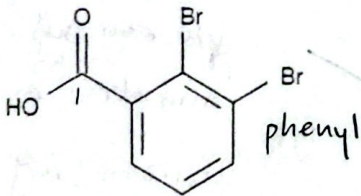
1. Complete the following table. Name the structural formula OR draw the structural formula corresponding to each name. Line drawings or structural formulas are acceptable. (10 marks)

a) o-diethylbenzene



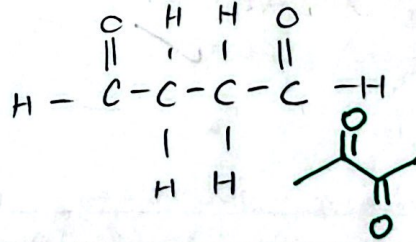
2-methylpropan-1-amine

c)

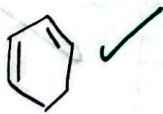


2,3-dibromophenylmethanoic acid

d) butanedione



e) cyclohexa-1,3-diene

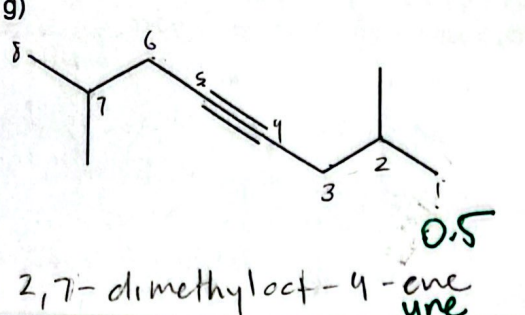
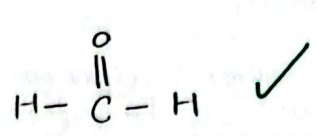
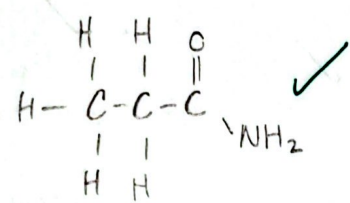
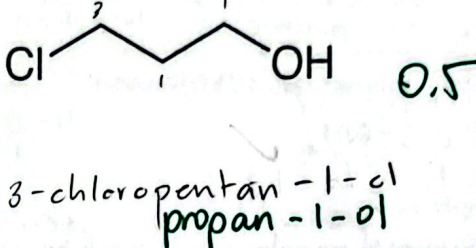


f) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{O} - \text{CH}_3$

methoxybutane

propanoic acid methyl is small so less non-polar parts

7

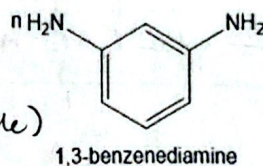
g)  2,7-dimethyloct-4-ene	h) formaldehyde meth 
i) propanamide 	j)  3-chloropentan-1-ol

Application (11 marks)

1. Many workers in hazardous environments—including race-car drivers—wear protective clothing made from a polymer called Nomex. Nomex is synthesized using the two monomers shown below. (4 marks total)

a) What type of polymerization reaction must occur to form Nomex? (1 mark)

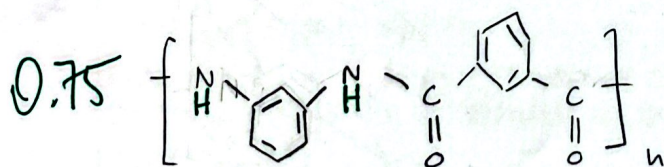
condensation (to form polyamide)



b) Will there be a by-product of the polymerization reaction? If so, what is the by-product? (1 mark)

water, HCl will be a by-product → HCl will be the by-product with this example

c) Draw a structural diagram showing one repeating unit of the polymer product. (1 mark)



d) What type of linkages form between the monomers? Because of these linkages, what is the name of the type of polymer called? (1 mark)

amide linkages form because of the carboxylic acid + NH₂ so the name of the type of polymer is polyamide

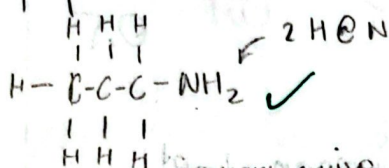
2. Amines can exist as primary, secondary, and tertiary amines. Examine the chart below and explain why their boiling points are so different. Use structural formulas to aid in your explanation. (3 marks)

Table 1: Various boiling points of isomers of C_3H_9N

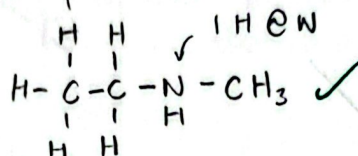
Compound	Boiling point ($^{\circ}C$)
Propan-1-amine	49
N-methylethan-1-amine	37
N,N-dimethylmethanamine	3

primary amines have a NH_2 group, allowing for H-bonding (so $\uparrow$ BP because more energy required to break forces), secondary amines have H-bonding too, but less strong than primary because only 1 H is attached to the nitrogen. tertiary amines cannot do H-bonding because there is no hydrogens attached to the nitrogen (so only dipole-dipole bc of N-C so lower BP)

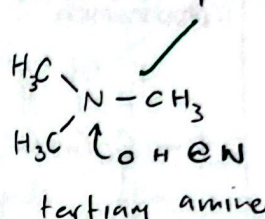
propan-1-amine



N-methylethan-1-amine

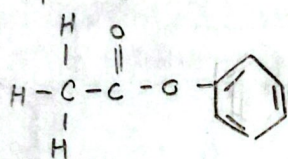


N,N-dimethylmethanamine

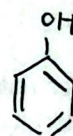
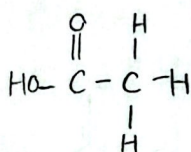


3. You want to create an ester but only have limited supplies. Show, using a series of reactions, how you could make phenyl ethanoate from appropriate alcohols. Name all reactants and products and use at least one (1) oxidation reaction. (4 marks)

phenyl ethanoate



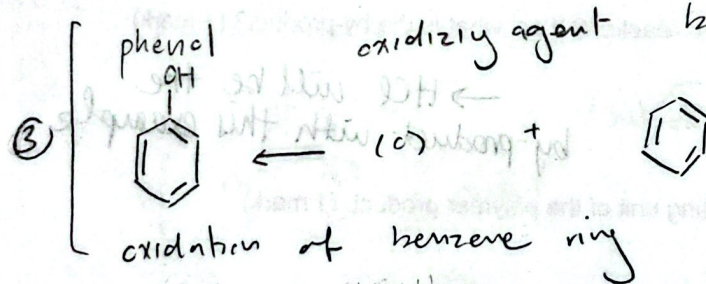
ethanoic acid + phenol



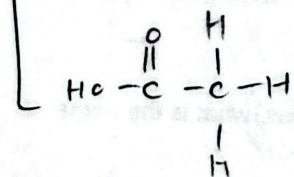
last step: esterification

benzene with phenol, no need for reaction

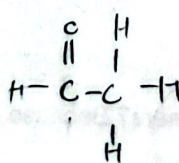
why are you writing reactions/steps backwards?



ethanoic acid



ethanal



+ oxidizing agent

